

Daniel Feitosa

Assistant Professor

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Employment

Assistant Professor Faculty of Science and Engineering, University of Groningen	Nov. 2021 - present
Assistant Professor Campus Fryslân, University of Groningen	Mar. 2019 - Oct. 2021
Guest Researcher Faculty of Science and Engineering, University of Groningen	Mar. 2019 - Oct. 2021
Lecturer Campus Fryslân, University of Groningen	Jun. 2018 - Feb. 2019
Researcher and Chief Data Scientist Data Research Centre, University of Groningen	Dec. 2017 - Feb. 2019
Researcher Faculty of Science and Engineering, University of Groningen	May 2017 - Dec. 2017
Software Engineer TQS Informática Ltda.	Aug. 2012 - Jan. 2013
Teaching Assistant (Undergraduate courses) University of São Paulo	Feb. 2007 - Jul. 2012
Research Assistant (National research projects) University of São Paulo	Feb. 2007 - Jul. 2012

Education

Ph.D. in Computer Science <i>University of Groningen</i> Thesis: Applying Patterns in Embedded Systems Design for Managing Quality Attributes and their Trade-offs Supervisors: Prof. Dr. Ir. Paris Avgeriou, Prof. Dr. Elisa Yumi Nakagawa, Dr. Apostolos Ampatzoglou	May 2013 - Jan. 2019
M.Sc. in Mathematics and Computational Sciences <i>University of São Paulo</i> Thesis: SiMuS - A Reference Architecture for Service Multirobotic Systems Supervisor: Prof. Dr. Elisa Yumi Nakagawa	Aug. 2010 - Feb. 2013
B.Sc. in Computer Science <i>University of São Paulo</i> Thesis: Software Engineering in Embedded Software and Mobile Robot Software Development: A Systematic Mapping Supervisor: Prof. Dr. Elisa Yumi Nakagawa	Feb. 2006 - Jul. 2010

Membership in Scientific & Professional Bodies

- **Journal of Systems and Software, Elsevier** 2025-present
Publicity Editor
- **Institute of Electrical and Electronics Engineers (IEEE)** 2018-present
IEEE Computer Society and IEEE Young Professionals
- **Dutch National Association for Software Engineering (VERSEN)** 2019-present
Leader of the Web Working Group (since Oct. 2021)
- **Steering Committee of European Conference on Software Architecture (ECSA)** 2023-present
Board support

Languages

Portuguese: Native English: Fluent Dutch: Basic (B1)

Professionalization

- **Academic Leadership I** May 2026 - Jun. 2026
University of Groningen (4 days)
- **Senior Teaching Qualification (STQ)** Feb. 2025 - Feb. 2026
University of Groningen
- **Inclusive Recruitment Training** Oct - Nov. 2024
University of Groningen (10 hours)
- **PhD Supervision Excellence** Apr. - Jun. 2024
University of Groningen (15 hours)
- **University Teaching Qualification (BKO)** Feb. 2023
University of Groningen
- **Training for Online Teaching and Online Exam** Mar. 2020
University of Groningen (1 day)
- **Training Program on Intercultural Competences** Mar. - Jun. 2018
University of Groningen (4 days)
- **Training Program on Teaching for PhD Students** Nov. - Dec. 2015
University of Groningen (4 days)
- **Training Program on Presentation Skills** Sep. 2015
Graduate School of Science (2 days)
- **Training for Thesis Supervision** Nov. 2014
University of Groningen (1 day)
- **Training Program on Publishing in English for PhD students** Sep. - Nov. 2013
University of Groningen (7 half-days)
- **CASE Summer School on Practical Experimentation in Software Engineering** Jul. 2013
Free University of Bozen-Bolzano, Italy
- **Training Program on Pedagogical Preparation** Aug. - Dec. 2010
University of São Paulo (60 hours)

Distinctions

- **2026 - Best Paper Award**
9th International Conference on Technical Debt (TechDebt 2025)
Paper: "Investigating CI/CD-based Technical Debt Management in Open-source Projects"
- **2026 - Distinguished Reviewer Award**
9th International Conference on Technical Debt (TechDebt 2025)
- **2025 - ACM SIGSOFT Distinguished Paper Award**
IEEE/ACM 47th International Conference on Software Engineering (ICSE 2025)
Paper: "PairSmell: A Novel Perspective Inspecting Software Modular Structure"
- **2025 - Distinguished Reviewer Award**
22th International Conference on Mining Software Repositories (MSR 2025)
- **2023 - ACM SIGSOFT Distinguished Paper Award**
2nd International ACM/IEEE International Conference on AI Engineering - Software Engineering for AI (CAIN 2023)
Paper: "Uncovering Energy-Efficient Practices in Deep Learning Training: Preliminary Steps Towards Green AI"
- **2023 - Distinguished Reviewer Award**
20th International Conference on Mining Software Repositories (MSR 2023)
- **2020 - 2nd place Teacher of Year**
Campus Fryslân, University of Groningen
- **2020 - Nominee for the GEC Best PhD Thesis Award**
Groningen Engineering Center
- **2010 - Early Access to Master's Program**
University of São Paulo
Due to outstanding academic performance
- **2009 - 1st place in Track 1 of the I Brazilian Computational Linguistics Olympiad**
Brazilian Computer Science Society

Teaching

Lecturing and Coordination

Master Thesis - Computing Science (M.Sc.) <i>Coordinator, University of Groningen</i>	Sep. 2025 - present
Machine Learning Systems Deployment and Operations (M.Sc.) <i>Lecturer and Coordinator, University of Groningen</i>	Nov. 2024 - present
Advanced Programming (B.Sc.) <i>Lecturer and Coordinator, University of Groningen</i>	Sep. 2024 - present
Advanced Object-Oriented Programming (B.Sc.) <i>Lecturer and Coordinator, University of Groningen</i>	Sep. 2022 - Oct. 2023
Algorithmic Programming Contests (B.Sc.) <i>Lecturer, University of Groningen</i>	Sep. 2022 - Jul. 2025
Problem Analysis and Software Design (B.Sc.) <i>Lecturer and Coordinator, University of Groningen</i>	Nov. 2021 - Jan. 2023
Software Architecture (M.Sc.) <i>Lecturer, University of Groningen</i>	Sep. 2021 - present

Topics on Data Science (B.Sc.) <i>Lecturer and Coordinator, Campus Fryslân</i>	Feb. 2021 - Oct. 2021
Living Lab (B.Sc.) <i>Supervisor, Campus Fryslân</i>	Feb. 2020 - Oct. 2021
Information Technology and its Implications (B.Sc.) <i>Lecturer and Coordinator, Campus Fryslân</i>	Oct. 2019 - Oct. 2021
Introduction to Data Science (B.Sc.) <i>Lecturer and Coordinator, Campus Fryslân</i>	Apr. 2019 - Oct. 2020
Introduction to Programming (B.Sc.) <i>Lecturer and Coordinator, Campus Fryslân</i>	Sep. 2018 - Oct. 2021
Portfolio - Global Responsibility and Leadership (B.Sc.) <i>Mentor, Campus Fryslân</i>	Sep. 2018 - Oct. 2021

Teaching Assistance

Software Patterns (M.Sc.) <i>University of Groningen</i>	Nov. 2014 - Jan. 2016
Information Systems (B.Sc.) <i>University of São Paulo</i>	Aug. 2011 - Jan. 2012
Analysis of Object-Oriented Designs (B.Sc.) <i>University of São Paulo</i>	Feb. 2010 - Jul. 2011
Databases Laboratory (B.Sc.) <i>University of São Paulo</i>	Aug. 2009 - Jan. 2010
Databases (B.Sc.) <i>University of São Paulo</i>	Feb. 2009 - Jul. 2009
Introduction to Computer Science (B.Sc.) <i>University of São Paulo</i>	Feb. 2007 - Jul. 2007

Supervision

PhD Candidates

Saagarika Gosu <i>Co-supervisor</i> Managing Technical Sustainability and its Tradeoffs	Aug. 2025 - present
Jesse Maarleveld <i>Co-supervisor</i> Machine Learning for Technical Debt Management	Dec. 2023 - present
João Paulo Biazotto <i>Co-supervisor</i> Technical Debt Management under Adverse Conditions	May 2019 - present
Eko Rahmadian <i>Co-supervisor – Defended</i> Governing the use of Big Data and Digital Twin Technology for Sustainable Tourism	Apr. 2019 - Mar. 2024
Jie Tan <i>Co-supervisor – Defended</i> Technical Debt Repayment in Practice	Feb. 2019 - Dec. 2021

Master Students

Alexander Zandman <i>Supervisor</i> A Testing Framework for RAG-based Applications	Feb. 2026 - Jul. 2026
Danny Jelsma <i>Co-Supervisor</i> Detecting and Repairing Infrastructure as Code Cost Antipatterns Using Large Language Models	Feb. 2026 - Jul. 2026
Allia Neamt <i>Supervisor</i> Benchmarking a Geospatial Foundational Model - Insights for Model Optimisation, Market Positioning and Internal Operational Improvements	Feb. 2026 - Jul. 2026
Denisa-Cristina Bunea <i>Supervisor</i> Development of RAG Modules for a Learning Assistant	Apr. 2026 - Jul. 2026
Ioannis Loukas <i>Supervisor</i> Design and Implementation of a Web Interface for a RAG Learning Assistant	Apr. 2026 - Jul. 2026
Alexandru-Traian Pennert <i>Supervisor</i> Cost Management and Cloud Footprint in a non-Trivial Proprietary System	Apr. 2026 - Jul. 2026
Alexander Zandman <i>Supervisor</i> Software Architecture & DevOps at Topicus	Apr. 2025 - Aug. 2025
Mojmír Majer <i>Co-Supervisor</i> Dynamic AI Code Optimization for Sustainable Software	Feb. 2025 - Feb. 2026
Robin Guichelaar <i>Co-Supervisor</i> Predicting cost-related actions on IaC artifacts using Machine Learning	Feb. 2025 - Aug. 2025
Somak Chatterjee <i>Co-Supervisor</i> Energy Behaviour in Workflow Applications	Sep. 2024 - Feb. 2025
Krishan Jokhan <i>Co-Supervisor</i> Assessing the environmental footprint of chatbot frameworks	Sep. 2024 - Mar. 2025
Andrei Dumitriu <i>Supervisor</i> A Digital Cybersecurity Compliance Evaluation Platform for CGI	Jan. 2025 - Jul. 2025
Germán Calcedo Pérez <i>Supervisor</i> Enhancing and Evaluating Language Models for AI-Assisted Study Selection	Feb. 2024 - Aug. 2024
Koen Bolhuis <i>Co-Supervisor</i> A linter for cost issues in cloud software	Feb. 2024 - Jul. 2024
Andrei Dumitriu <i>Supervisor</i> Upgrade design pattern detector and quality assessment	Apr. 2023 - Jun. 2024

Túpac Rocher <i>Supervisor</i> A student-friendly continuous integration pipeline	Jun. 2022 - Aug. 2022
Tim Yarally <i>Co-supervisor</i> Assessing the energy efficiency of training and execution of machine learning models	Nov. 2021 - Jul. 2022
Eric Rwemigabo <i>Co-supervisor</i> Integrating a design pattern detector into SEAGle	May 2017 - Jul. 2017
Rutger Alders <i>Co-supervisor</i> Trade-offs in performance and energy consumption of design patterns	May 2015 - Feb. 2016
Jeroen David van Leusen <i>Co-supervisor</i> Static analyzer for C/C++	Apr. 2015 - Jun. 2015

Bachelor Students

Alexandra Gabriela Vasile and Bianca Sabina Iliuta <i>Supervisor (Thesis)</i> AI-Assisted Development and Cost-Awareness Evaluation of a Cloud-Native Benchmark Application	Feb. 2026 - Jul. 2026
Ioana-Teodora Stanca <i>Supervisor (Internship)</i> Production-Ready Case Instance Interface at Klippa	Sep. 2025 - Dec. 2025
Ollie Eilon <i>Co-supervisor (Thesis)</i> Technology Research and Redesign in Rust for Enhanced Efficiency	Feb. 2025 - Jul. 2025
Denisa-Cristina Bunea <i>Co-supervisor (Thesis)</i> Technology Research and Redesign in Rust for Enhanced Maintainability and Developer Efficiency	Feb. 2025 - Jul. 2025
Vlad Ichim <i>Supervisor (Thesis)</i> Investigating cost-related changes in Infrastructure as Code	Feb. 2025 - Jul. 2025
Mattia Marziali <i>Supervisor (Thesis)</i> Supporting software repository mining for empirical research	Feb. 2025 - Jul. 2025
Vlad Ichim <i>Supervisor (Honours College Project)</i> Analyzing Cherry-Picking Practices in Open-Source Software Development	Nov. 2024 - May. 2025
Mattia Marziali <i>Supervisor (Short Programming Project)</i> Data scraping and processing at scale	Nov. 2024 - Jan. 2025
Madalina Ioana Gavat <i>Supervisor (Thesis)</i> Technical Debt detection using Graph Neural Networks	Feb. 2024 - Jul. 2024
Andrei S. Istudor <i>Supervisor (Thesis)</i> Applying Graph Learning For Technical Debt Detection	Feb. 2024 - Jul. 2024

Fadziso Manwa <i>Supervisor</i> (Thesis) Identifying requirements for technical debt tools	Feb. 2024 - Jul. 2024
Tudor Roscoiu <i>Supervisor</i> (Thesis) TD Bot: A bot for supporting technical debt management in GitHub issues	Feb. 2024 - Jul. 2024
Andrew Rutherford <i>Co-supervisor</i> (Thesis) NeoRepro: A Tool For Creating Replication Packages	Feb. 2024 - Jul. 2024
Nicolae Tkacenko <i>Co-supervisor</i> (Thesis) Investigating Cost Awareness in Amazon CloudFormation	Feb. 2024 - Aug. 2024
Alexandru-Nicolae Cirlan <i>Co-supervisor</i> (Thesis) Mining for Cost Awareness in Cloud Computing	Feb. 2024 - Aug. 2024
Allia I. Neamt <i>Co-supervisor</i> (Thesis) From Terraform to AWS CloudFormation: A Study of Cost Patterns and Antipatterns	Feb. 2024 - Jul. 2024
Abel A. van der Til <i>Co-supervisor</i> (Thesis) Cost awareness in pull requests of open source repositories	Feb. 2024 - Aug. 2024
Andrei-Stefan Istudor <i>Supervisor</i> (Internship) Software Mining Rig: Building a Scalable MSR Infrastructure	Dec. 2023 - Jun. 2024
Oscar de Francesca and Konstantinos Chasiotis <i>Supervisor</i> (Short Programming Project) Bot for technical debt detection in issue tracking systems	Nov. 2023 - Mar. 2024
Khanak Gulati <i>Supervisor</i> (Thesis) Empirical Evaluation of ChatGPT's Ability to Solve Coding Problems	Feb. 2023 - Aug. 2023
Ana Terna and Karol Machnik <i>Supervisor</i> (Thesis) Is Design Pattern Grime Related to Technical Debt?	Feb. 2023 - Jul. 2023
Mohammad Al Shakoush <i>Co-supervisor</i> (Thesis) Empirical Sentiment Analysis for Predicting Cost Management Actions	Feb. 2023 - Aug. 2023
Kasper Johannes Hopen Alfarnes <i>Supervisor</i> (Short Programming Project) SEARCH website	Feb. 2023 - Jun. 2023
Andrei-Stefan Istudor <i>Supervisor</i> (Honours College Project) Software engineering and architecture for automotive resource-limited embedded systems	Jan. 2023 - Jun. 2024
Andrei Valentin Girjoaba <i>Co-supervisor</i> (Honours College Project) Natural deduction proof evaluator for Fitch style proofs	Nov. 2022 - Sep. 2023

Mohammad Al Shakoush <i>Co-supervisor</i> (Short Programming Project) Distributed machine-learning pipeline for software engineering sentiment analysis	Nov. 2022 - Apr. 2023
Tudor Dragan <i>Supervisor</i> (Honours College Project) IoT and the Mirai malware	Jul. 2022 - Aug. 2023
Massimiliano Berardi, Rares-Dorian Boza and Matei-Tudor Penca <i>Co-supervisor</i> (Thesis) Mining and analysis of cost-related decisions in cloud infrastructures	Mar. 2022 - Jul. 2022
Jana Heitkemper <i>Supervisor</i> (Internship) Netwoke: improving networking in closed environments	May 2020 - Jan. 2021
Abdussamet Dumankaya <i>Supervisor</i> (Internship) Automation of Quality Analysis of Java Projects	Jul. 2016 - Sep. 2016
Marijn Scholtens <i>Supervisor</i> (Internship) Improving Quality Analysis during Continuous Integration	Jan. 2016 - Jul. 2016
Begum Benel and Can Berker Çıkış <i>Supervisor</i> (Internship) Static analyzer for C/C++	Jun. 2015 - Aug. 2015
Çağla Koca and Melda Seren İnan <i>Supervisor</i> (Internship) Summer internship projects	Jun. 2014 - Sep. 2014

Technical Contributions

TagDebt Bot GitHub bot for supporting Technical Debt Management using machine learning. https://github.com/search-rug/tagdebt-bot	Python
Student-friendly Report GitHub action to provide feedback on Java source code quality. https://github.com/search-rug/student-friendly-report	Javascript
SSAP Tool for detecting extended pattern participants in GoF design patterns. https://github.com/search-rug/ssap	Scala
Spoon-pttgrime Tool for calculating pattern grime metrics of GoF pattern instances. https://github.com/search-rug/spoon-pttgrime	Java
SEagle Tool for software evolution analysis. https://bitbucket.org/feitosa-daniel/seagle-server/src	Java
Cpptool C/C++ static analyzer. https://github.com/search-rug	C++, Java

Academic Services

Member of the Computer Science Curriculum Committee <i>University of Groningen</i>	Nov. 2025 - present
Member of the Computer Science Program Committee <i>University of Groningen</i>	Nov. 2021 - 2025
Curriculum developer for B.Sc. Data Science and Digital Society <i>Campus Fryslân</i>	Jun. 2020 - Nov. 2021
Member of the Admissions Board <i>University College Fryslân</i>	Jun. 2018 - Oct. 2021
Curriculum developer for B.Sc. Global Responsibility and Leadership <i>Campus Fryslân</i>	Feb. 2017 - Aug. 2017

Professional Services

Organization of Scientific Events

- **Program Co-Chair**
10th International Conference on Technical Debt (TechDebt '27)
- **Co-Organizer**
International Software Architecture PhD School (ISAPS '23 - '26)
- **Co-Chair Poster track**
23rd IEEE International Conference on Software Architecture (ICSA '26)
- **Co-Organizer**
School of Empirical Studies in Software Engineering (EES '26)
- **Co-Chair Green and Sustainable Computing track**
18th International Conference on the Quality of Information and Communications Technology (QUATIC '25)
- **Co-Chair Junior PC**
8th International Conference on Technical Debt (TechDebt '25)
- **Co-Chair Software Engineering and Debt Metaphors (SEaDeM) track**
Euromicro Conference Series on Software Engineering and Advanced Applications (SEAA '22 - '23)
- **Virtualization Chair**
29th International Conference on Software Analysis, Evolution and Reengineering (SANER '22)
- **Web Chair**
4th International Conference on Technical Debt (TechDebt '21)
- **Co-Organizer**
5th International Workshop on Machine Learning Techniques for Software Quality Evaluation (MaLTesQuE '21)
- **Chair - track Journal 1st**
46th Euromicro Conference Series on Software Engineering and Advanced Applications (SEAA '20)

Scientific Reviews

Journals:

- ACM Computing Surveys (CSUR)
- Communications of the ACM. (CACM)
- Empirical Software Engineering (EMSE)

- IEEE Access
- IEEE Computer
- IEEE Software
- IEEE Transactions on Software Engineering (TSE)
- ACM Transactions on Software Engineering and Methodology (TOSEM)
- IET Software
- Information and Software Technology (IST)
- International Journal of Network Management (NEM)
- Journal of Systems and Software (JSS)
- Journal of Software: Evolution and Process (JSME)
- Journal of Computer Science and Technology (JCST)

Conferences, Workshops and Books:

- International Conference on Software Engineering (ICSE 2016, 2017, 2023, 2025–2027)
- International Conference on Source Code Analysis and Manipulation (SCAM 2026)
- International Conference on Mining Software Repositories (MSR 2023–2026)
- Empirical Software Engineering and Measurement (ESEM 2016, 2017, 2022, 2025)
- International Conference on Software Architecture (ICSA 2017, 2018, 2022, 2023, 2025)
- International Conference on Technical Debt (TechDebt 2023–2026)
- International Conference on Software Maintenance and Evolution (ICSME 2024, 2025)
- International Conference on the Quality of Information and Communications Technology (QUATIC, 2024, 2025)
- ICT Open (2025)
- International Conference on Software Analysis, Evolution and Reengineering (SANER 2024, 2026)
- Euromicro Conference on Software Engineering and Advanced Applications (SEAA 2020–2026)
- Brazilian Symposium on Software Engineering (SBES 2016, 2021–2024)
- International Conference on Program Comprehension (ICPC 2022, 2023)
- Symposium on Applied Computing (SAC 2022)
- Workshop on Software Quality Assurance for Artificial Intelligence (SQA4AI 2022)
- International Workshop on Technical Debt for Variability-intensive Systems (TD4Vis 2022)
- European Conference on Software Architecture (ECSA 2015, 2017, 2018, 2020, 2021)
- Machine Learning Techniques for Software Quality Evaluation Workshop (MaLTesQuE 2019–2021)
- International Conference on Software Reuse (ICSR 2016, 2018)
- Evaluation and Assessment in Software Engineering Conference (EASE 2016, 2017, 2018)
- International Conference on Advanced Information Systems Engineering (CAiSE 2015, 2016)
- Australasian Software Engineering Conference (ASWEC 2015)
- Working Conference on Software Architecture (WICSA 2014, 2015)

- Brazilian Symposium on Software Components, Architectures, and Reuse (SBCARS 2014)
- International track on adaptive and reconfigurable service-oriented and component-based Applications and Architectures (AROSA 2014)
- International Conference on Model-Driven Engineering and Software Development (MODELSWARD 2014)
- International Workshop on Software Engineering for Resilient Systems (SERENE 2013)

Expert Evaluation for Funding Agencies

- Fundação de Amparo à Ciência e Tecnologia de Pernambuco, Brazil (FACEPE, 2024)

Funding

- **2025 – NRO Comenius Teaching Fellow grant** €50,000.00
A Students-First Approach to Developing AI-based Systems
- **2024 – ACM SIGSOFT Grant for Summer/Winter Schools** \$5,000.00
International Software Architecture PhD School
- **2022 – RuG PhD4eTT** approx. €280,000 (1 PhD position)
Fusing Data Sources for Monitoring, Prioritizing, and Predicting Technical Debt Hotspots

Publications

Journals

- J1 C. Zhong, D. Feitosa, P. Avgeriou, H. Huang, W. Song, H. Zhang. "What Characterizes Pairwise Modular Smells?". *IEEE Transactions on Software Engineering*, in press, 2026.
- J2 L. Carvalho, J.P. Biazotto, D. Feitosa, R. Kazman, E.Y. Nakagawa. "Technical debt management in continuous software engineering: State of the art and research agenda". *Journal of Software: Evolution and Process*, in press, 2026.
- J3 J.P. Biazotto, D. Feitosa, P. Avgeriou, E.Y. Nakagawa. "TagDebt: A Bot to Support Technical Debt Management". *Empirical Software Engineering*, 2026. DOI: 10.1007/s10664-026-10900-9.
- J4 K. Bolhuis, A. Neamt, D. Feitosa, V. Andrikopoulos. "Towards sustainable cloud deployments: A cost (anti)patterns catalog for terraform and CloudFormation". *Journal of Systems and Software*, 2026. DOI: 10.1016/j.jss.2026.112870.
- J5 J. Maarleveld, J. Guo, D. Feitosa. "A systematic mapping study on graph machine learning for static source code analysis". *Information and Software Technology*, 2025. DOI: 10.1016/j.infsof.2025.107722.
- J6 J.P. Biazotto, D. Feitosa, P. Avgeriou, E.Y. Nakagawa. "Understanding practitioners' reasoning and requirements for efficient tool support in technical debt management". *Empirical Software Engineering*, 2025. DOI: 10.1007/s10664-025-10691-5.
- J7 L. Cruz, J.P. Fernandes, M.H. Kirkeby, S. Martínez-Fernández, J. Sallou, H. Anwar, E. Barba Roque, J. Bogner, J. Castaño, F. Castor, A. Chasmawala, S. Cunha, D. Feitosa, A. González, A. Jedlitschka, P. Lago, H. Muccini, A. Oprescu, P. Rani, J. Saraiva, F. Sarro, R. Selvan, K. Vaidhyanathan, R. Verdecchia, I.P. Yamshchikov. "Greening AI-enabled Systems with Software Engineering: A Research Agenda for Environmentally Sustainable AI Practices". *ACM SIGSOFT Software Engineering Notes*, 2025. DOI: 10.1145/3743095.3743099.
- J8 D. Feitosa, M. Penca, M. Berardi, R. Boza, V. Andrikopoulos. "Mining for cost awareness in the infrastructure as code artifacts of cloud-based applications: An exploratory study". *Journal of Systems and Software*, 2024. DOI: 10.1016/j.jss.2024.112112.

- J9 J.P. Biazotto, D. Feitosa, P. Avgeriou, E.Y. Nakagawa. "Technical debt management automation: State of the art and future perspectives". *Information and Software Technology*, 2024. DOI: 10.1016/j.infsof.2023.107375.
- J10 N. Nikolaidis, N. Mittas, A. Ampatzoglou, D. Feitosa, A. Chatzigeorgiou. "A metrics-based approach for selecting among various refactoring candidates". *Empirical Software Engineering*, 2024. DOI: 10.1007/s10664-023-10412-w.
- J11 N. Nikolaidis, E. Arvanitou, C. Volioti, T. Maikantis, A. Ampatzoglou, D. Feitosa, A. Chatzigeorgiou, P. Krief. "Eclipse Open SmartCLIDE: An end-to-end framework for facilitating service reuse in cloud development". *Journal of Systems and Software*, 2024. DOI: 10.1016/j.jss.2023.111877.
- J12 J. Tan, D. Feitosa, P. Avgeriou. "The lifecycle of Technical Debt that manifests in both source code and issue trackers". *Information and Software Technology*, 2023. DOI: 10.1016/j.infsof.2023.107216.
- J13 C. Zhong, H. Zhang, C. Li, H. Huang, D. Feitosa. "On measuring coupling between microservices". *Journal of Systems and Software*, 2023. DOI: 10.1016/j.jss.2023.111670.
- J14 E. Rahmadian, D. Feitosa, Y. Virantina. "Digital twins, big data governance, and sustainable tourism". *Ethics and Information Technology*, 2023. DOI: 10.1007/s10676-023-09730-w.
- J15 J. Tan, D. Feitosa, P. Avgeriou. "Does it matter who pays back Technical Debt? An empirical study of self-fixed TD". *Information and Software Technology*, 2022. DOI: 10.1016/j.infsof.2021.106738.
- J16 E. Rahmadian, D. Feitosa, A. Zwitter. "A systematic literature review on the use of big data for sustainable tourism". *Current Issues in Tourism*, 2022. DOI: 10.1080/13683500.2021.1974358.
- J17 J. Tan, D. Feitosa, P. Avgeriou, M. Lungu. "Evolution of technical debt remediation in Python: A case study on the Apache Software Ecosystem". *Journal of Software: Evolution and Process*, 2021. DOI: 10.1002/smr.2319.
- J18 A. Gkortzis, D. Feitosa, D. Spinellis. "Software reuse cuts both ways: An empirical analysis of its relationship with security vulnerabilities". *Journal of Systems and Software*, 2021. DOI: 10.1016/j.jss.2020.110653.
- J19 D. Feitosa, A. Ampatzoglou, A. Gkortzis, S. Bibi, A. Chatzigeorgiou. "CODE reuse in practice: Benefiting or harming technical debt". *Journal of Systems and Software*, 2020. DOI: 10.1016/j.jss.2020.110618.
- J20 P. Smiari, S. Bibi, D. Feitosa. "Examining the reuse potentials of IoT application frameworks". *Journal of Systems and Software*, 2020. DOI: 10.1016/j.jss.2020.110706.
- J21 D. Feitosa, A. Ampatzoglou, P. Avgeriou, A. Chatzigeorgiou, E. Nakagawa. "What can violations of good practices tell about the relationship between GoF patterns and run-time quality attributes?". *Information and Software Technology*, 2019. DOI: 10.1016/j.infsof.2018.07.014.
- J22 D. Feitosa, A. Ampatzoglou, P. Avgeriou, E.Y. Nakagawa. "Correlating Pattern Grime and Quality Attributes". *IEEE Access*, 2018. DOI: 10.1109/access.2018.2829895.
- J23 D. Feitosa, R. Alders, A. Ampatzoglou, P. Avgeriou, E.Y. Nakagawa. "Investigating the effect of design patterns on energy consumption". *Journal of Software: Evolution and Process*, 2017. DOI: 10.1002/smr.1851.

International Conferences

- C1 S. Yoshimoto, K. Horikawa, D. Feitosa, Y. Kashiwa, H. Iida. "Is Self-Admitted Technical Debt Tested? An Empirical Study of Coverage, Co-change, and Impact". In *Proceedings of the 2026 International Symposium on Empirical Software Engineering and Measurement (ESEM '26)*, 2026.
- C2 A. Gupta, R. Dhar, D. Feitosa, K. Vaidhyanathan. "Context Matters: Evaluating Context Strategies for Automated ADR Generation Using LLMs". In *Proceedings of the 30th International Conference on Evaluation and Assessment in Software Engineering (EASE '26)*, 2026. DOI: 10.1145/3816483.3816523.
- C3 João Paulo Biazotto, Daniel Feitosa, Paris Avgeriou, Elisa Yumi Nakagawa. "Investigating CI/CD-based Technical Debt Management in Open-source Projects". In *Proceedings of the 9th International Conference on Technical Debt (TechDebt '26)*, 2026. DOI: 10.1145/3794915.3795782. **(Distinguished Paper Award)**

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